1. DFA – String starts with a and ends with a

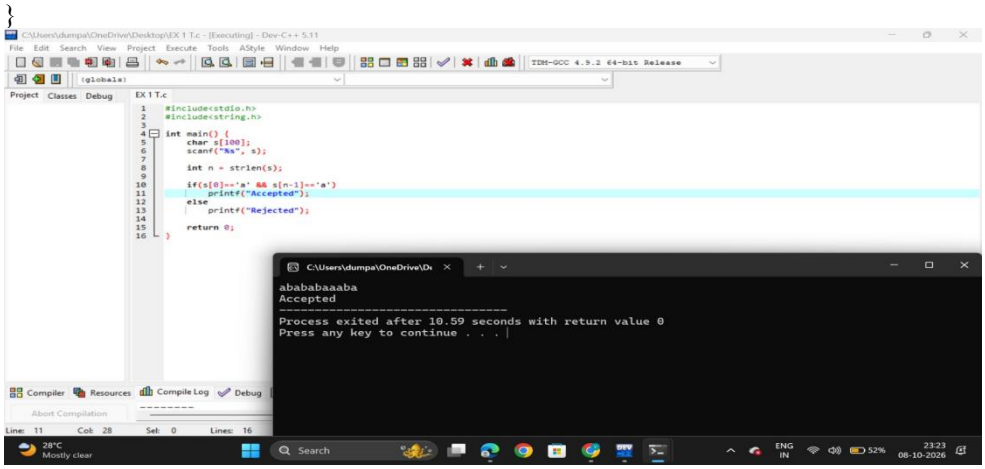
```
#include<stdio.h>
#include<string.h>

int main() {
   char s[100];
   scanf("%s", s);

   int n = strlen(s);

   if(s[0]=='a' && s[n-1]=='a')
      printf("Accepted");
   else
      printf("Rejected");

   return 0;
}
```



2. DFA – String starts with 0 and ends with 1

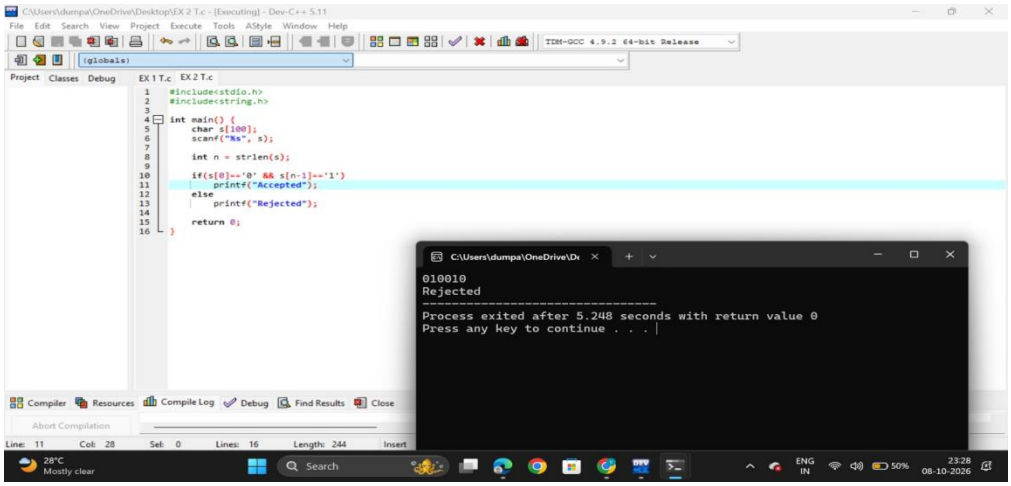
```
   #include<stdio.h>
#include<string.h>

int main() {
   char s[100];
   scanf("%s", s);

   int n = strlen(s);

   if(s[0]=='0' && s[n-1]=='1')
      printf("Accepted");
   else
      printf("Rejected");

   return 0;
}
```



3. CFG

```
#include <stdio.h>
#include <string.h>

int main() {
   char s[100];
   int i, n, valid = 1;

   printf("Enter a string: ");
   scanf("%99s", s);

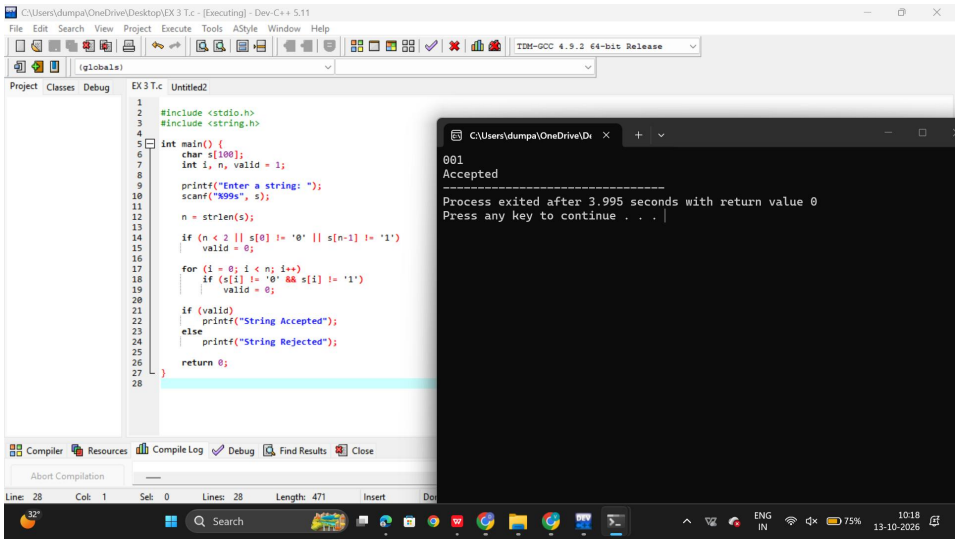
   n = strlen(s);

   if (n < 2 || s[0] != '0' || s[n-1] != '1')
      valid = 0;

   for (i = 0; i < n; i++)
      if (s[i] != '0' && s[i] != '1')
         valid = 0;

   if (valid)
      printf("String Accepted");
   else
      printf("String Rejected");

   return 0;
}
```



4. CFG

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, n, valid = 1;

    printf("Enter a string: ");
    scanf("%99s", s);

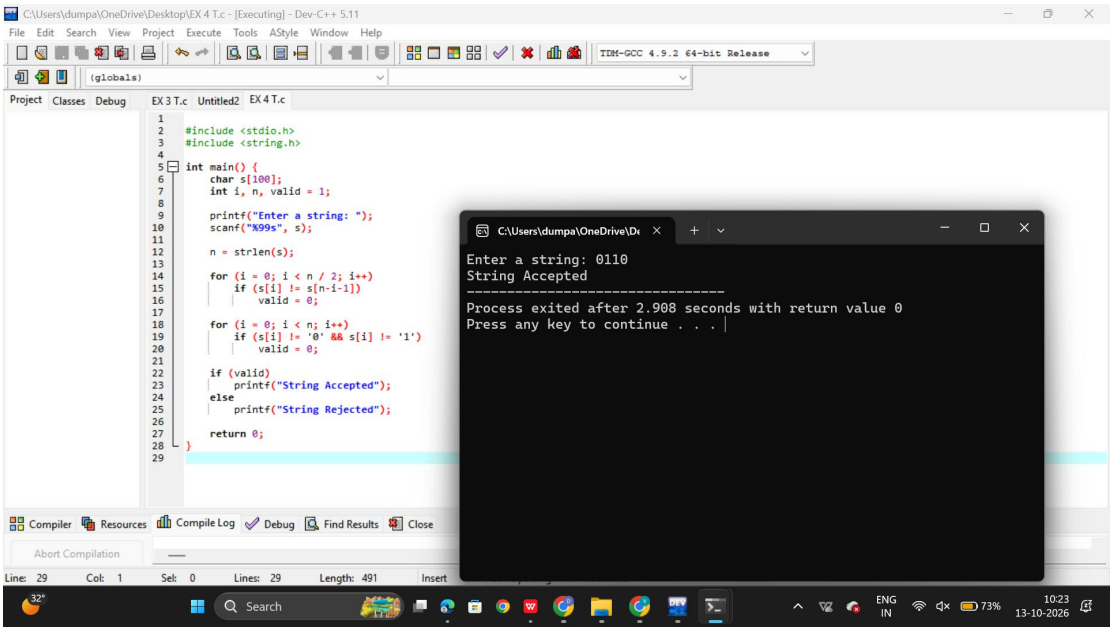
    n = strlen(s);

    for (i = 0; i < n / 2; i++)
        if (s[i] != s[n-i-1])
            valid = 0;

    for (i = 0; i < n; i++)
        if (s[i] != '0' && s[i] != '1')
            valid = 0;

    if (valid)
        printf("String Accepted");
    else
        printf("String Rejected");

    return 0;
}
```



5. CFG

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    int i, j, valid = 1;

    printf("Enter a string: ");
    scanf("%99s", s);

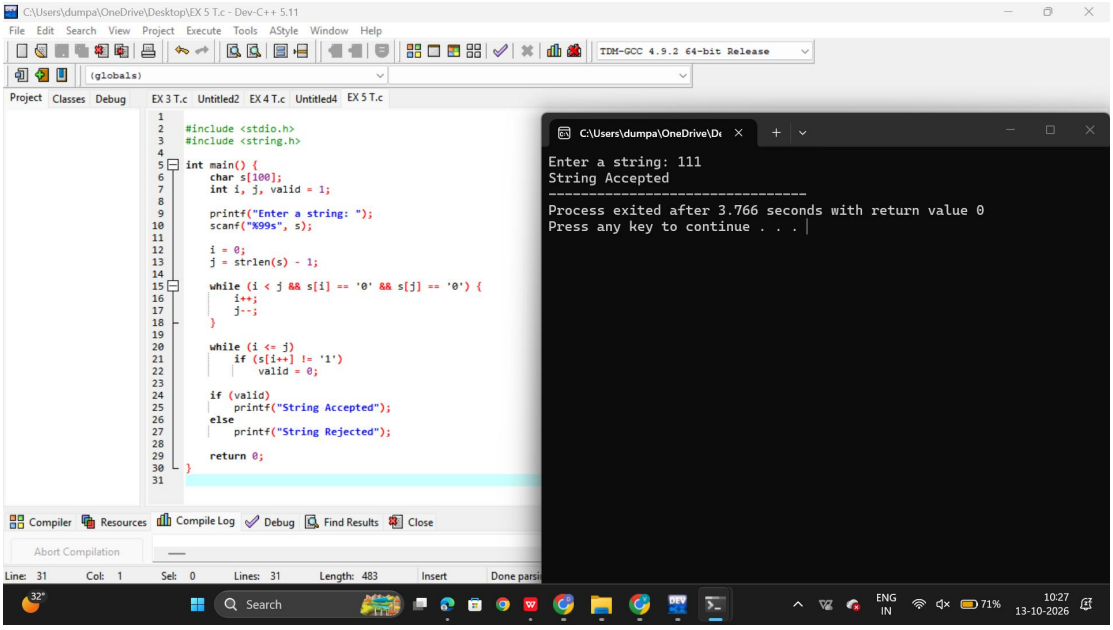
    i = 0;
    j = strlen(s) - 1;

    while (i < j && s[i] == '0' && s[j] == '0') {
        i++;
        j--;
    }

    while (i <= j)
        if (s[i++] != '1')
            valid = 0;

    if (valid)
        printf("String Accepted");
    else
        printf("String Rejected");

    return 0;
}
```


C:\Users\dumpa\OneDrive\Desktop\EX 5 T.c - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
EX 3 T.c Untitled2 EX 4 T.c Untitled4 EX 5 T.c
#include <stdio.h>
#include <string.h>
int main() {
char s[100];
int i, j, valid = 1;
printf("Enter a string: ");
scanf("%99s", s);
i = 0;
j = strlen(s) - 1;
while (i < j && s[i] == '0' && s[j] == '0') {
i++;
j--;
}
while (i <= j)
if (s[i++] != '1')
valid = 0;
if (valid)
printf("String Accepted");
else
printf("String Rejected");
return 0;
}
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Line: 31 Col: 1 Sel: 0 Lines: 31 Length: 483 Insert
C:\Users\dumpa\OneDrive\D
Enter a string: 111
String Accepted
Process exited after 3.766 seconds with return value 0
Press any key to continue . . .
Search
ENG
IN
71%
10:27
13-10-2026